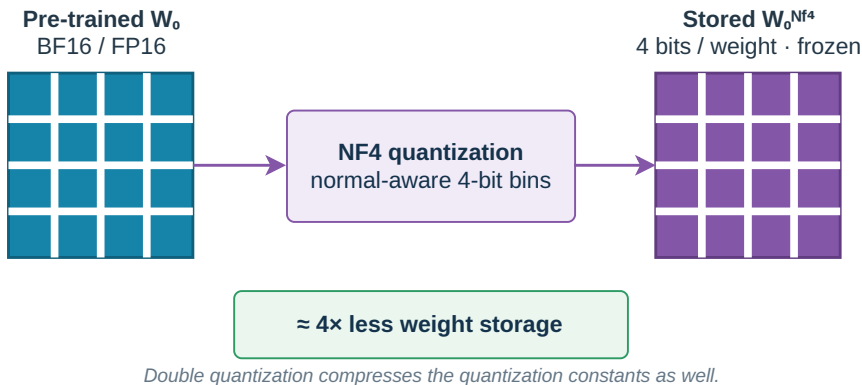
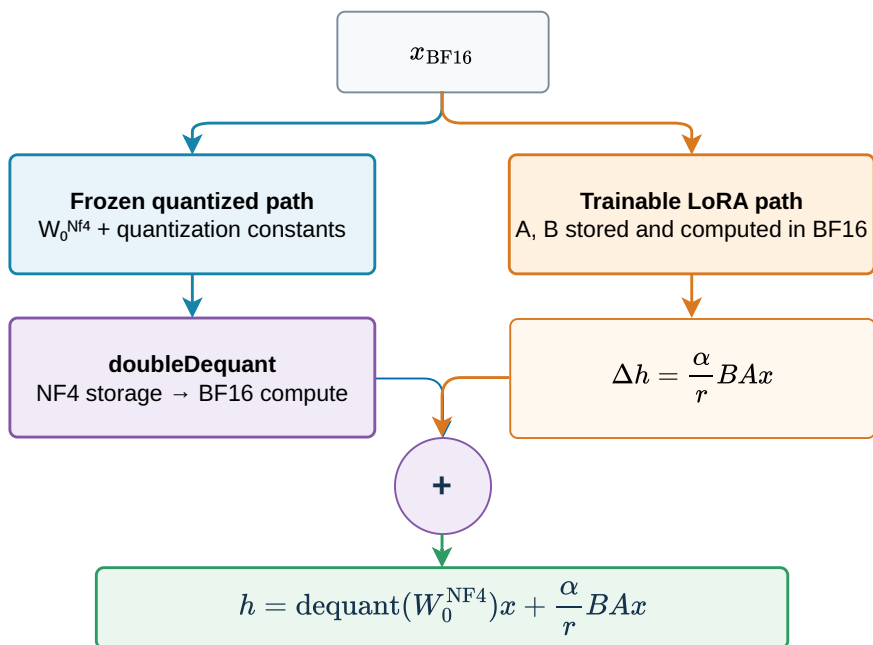


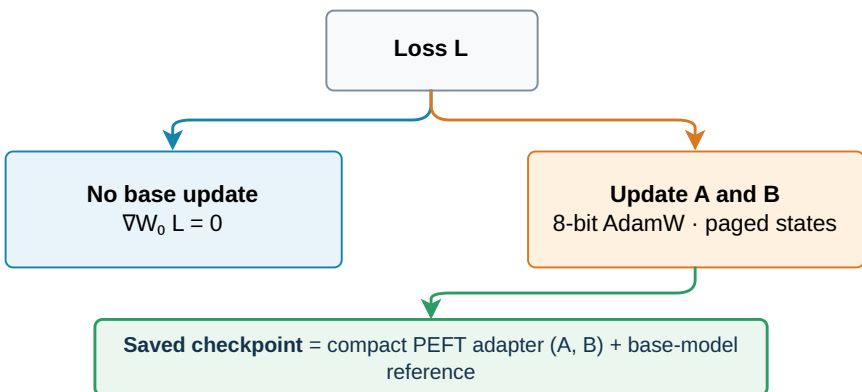
01 QUANTIZE THE FROZEN BACKBONE



02 QLoRA FORWARD PASS



03 BACKPROPAGATION AND SAVED ARTIFACT



QLoRA = 4-bit frozen backbone + BF16 LoRA adapters + gradients only through A and B